

Experiment No. 2 Data Cleaning

Aim:

- a) Handle missing values in a dataset using techniques like imputation or deletion.
- b) Identify and handle outliers using methods such as z-score or interquartile range.

Steps for -

Part a) Handle missing values in a dataset using techniques like imputation or deletion.

1. Read dataset : `train_u6lujuX_CVtuZ9i.csv` (will be provided during session)
2. Display its information and statistical summary.
3. Check for the missing values.`[df.isnull().sum()]`
4. **Methods to handle missing values in a dataset**

a. Deleting the Missing Values

1. **Deleting the row** with missing data : delete rows which have at least one NULL value. Even though other values are non null we delete the entire row if there is at least one NULL value. For instance, if every row has some (column) value missing, you might end up deleting the whole data.

E.g. `df.dropna(axis=0)`

2. **Deleting the column** with missing data : if the majority of the data points has NULL value for a particular column then we can just simply drop the entire column.

E.g. `df.drop(['Dependents'],axis=1)`

b. Imputing the Missing Values

1. Imputing missing values with mean/median

Columns in the dataset which are having numeric continuous values can be replaced with the mean, median, or mode of remaining values in the column.

This approach is popularly used when there are a small number of missing values in the data. However, when there are many missing values, mean or median results can result in a loss of variation in the data.

Mean imputation can provide a good estimate of the missing values for normally distributed data (the mean imputation is sensitive to outliers), and median for skewed data. It's better to use the median value for imputation in the case of outliers.

Use the 'fillna' method for imputing the columns 'LoanAmount' and 'Credit_History' with the mean of the respective column values.

```
df['LoanAmount'] = df['LoanAmount'].fillna(df['LoanAmount'].mean())
```

Use the 'fillna' method for imputing the column `Loan_Amount_Term` with the median of it.

2. Imputation method for categorical columns

When missing values are from categorical columns (string or numerical) then the missing values can be replaced with the most frequent category.

use the 'fillna' method for imputing the categorical columns 'Gender,' 'Married,' and 'Self_Employed.'

```
E.g. for Gender: df['Gender'].value_counts()
null_rows = df.query('Gender != Gender')
print(null_rows)
df['Gender'] = df['Gender'].fillna(df['Gender'].mode()[0])
print(df['Gender'].iloc[477])
```

Part b) Identify and handle outliers using methods such as z-score or interquartile range.

some of the techniques of detecting outliers

- Box Plots
- Z-score
- Interquartile Range(IQR)

1. load the given dataset:

https://drive.google.com/file/d/1dAwQLGJIfzfoSL45Zt6y5K_ctda-NSQm/view?source=post_page-----be8471a021a9-----

2. Visualize distribution of columns salary and salary_in_usd using boxplot. In the boxplots, **outliers** typically appear as points that lie outside the “whiskers” of the boxplot.
3. Statistical Methods to Detect Outliers:

a. Using IQR (Interquartile Range) :

The **IQR** method identifies outliers by calculating the range between the 25th percentile (Q1) and the 75th percentile (Q3) of the data. Any data point outside the range defined by **1.5 * IQR** (below Q1 or above Q3) is considered an outlier. **For example**, outside 1.5 times the interquartile range above the upper quartile and below the lower quartile ($Q1 - 1.5 * IQR$ or $Q3 + 1.5 * IQR$).

Apply following on salary_in_usd data column :

Calculate Q1,Q3 and IQR. Using above formula find upperbound and lowerbound
Compare your data points in salary_in_usd with bounds and display your outliers.
`outliers = df[(df['salary_in_usd'] < lower_bound) | (df['salary_in_usd'] > upper_bound)]`

b. Using Z-Score :

Another method for identifying outliers is using the **Z-score**, which measures how many standard deviations a data point is from the mean. Data points with a Z-score greater than **3** or less than **-3** are typically considered outliers.

Import: `from scipy import stats`

Calculate z-score for 'salary_in_usd' with `stats.zscore(your column)`
Compare these scores with 3 and -3 and print outliers.

4. Handle Outliers

a. Remove Outliers:

If the outliers are not critical to the analysis and seem to be errors or anomalies, we can remove them from the dataset.

Create a dataframe without outlier i.e., values greater than lower bound and less upper bound.

b. Impute Outliers :

we can impute outliers using the **mean**, **median**, or **mode**. This approach is often used when removing outliers would result in a significant loss of data.

5. Verify the Handling of Outliers :

You can do this by re-checking the distribution using boxplots or applying the IQR and Z-score methods again. Use boxplot to showcase removal of outliers.

Note:1. Handle all columns in dataset with outliers and

2. Also handle outliers in part(a) dataset